Assignment No-4

* Time Complexity & Space Complexity :

1)
```
* * * *
* * * *
* * * *
* * * *
```

```
int n=4;
for(int i=1; i<=n; i++){
    for(int j=1; j<=n; j++){
        Sop("*");
    }
    Soprintln();
```

i चा loop = 1 to n — n times
j चा loop = 1 to n — n times
$n*n = n^2$

1) Time Complexity $= n^2 = O(n^2)$
2) Space Complexity $= O(1)$

---

2)
```
*
* *
* * *
* * * *
```

```
int n=4;
for(int i=1; i<=n; i++){
    for(int j=1; j<=i; j++){
        Sop("*");
    }
}
```

i=1, j=1
i=2, j=1,2
i=3, j=1,2,3
i=4, j=1;2;3,4

j हा i पर्यंत फिरतोय. यासाठी sum of n natural no :-

$$\frac{n(n+1)}{2} = \frac{n^2+n}{2} = \frac{n^2}{2} = n^2 \quad O(n^2)$$

Time complexity $= O(n)^2$
Space Complexity $= O(1)$

3)

```
* * * *
* * *
* *
*
```

```
int n=4
for(int i=1; i<=n;i++){
    for(int j=1; j<=n;j++){
        if(i+j<=n+1){
            Sop("*");
        }else{
            Sop("-");
        }
    }
```

i चा loop 1 to n
j चा loop 1 to n.

for i
initialization = 1
Cond = n+1
increment = n
~~Print~~ = काम = n

= for j = init = 1
Cond = n+1
incre = n
print = n

i=1 j=1
i=2 j=1,2
i=3 j=1,2,3
i=4 j=1,2,3,4

Sum of n natural no:
$\frac{n(n+1)}{2} = \frac{n^2+n}{2}$
$= \frac{n^2}{2} = n^2$

time Complexity = $O(n^2)$

---

4)

```
* * * *
*   *
*   *
* * * *
```

```
int n=4;
for(int i=1;i<=n;i++){
    for(int j=1;j<=n;j++){
        if(i==1 || i==n || j==1 ||
            j==n){
            Sop("*");
        }else{
            Sop(" ");
        }
```

5)

i ~~cll~~ loop = 1 to n
j = loop = 1 to n

$n \times n = n^2$

time Complexity = $O(n^2)$
Space Complexity = $O(1)$

5)

```
      1
    1   2
  1   2   3
1   2   3   4
  1   2   3
    1   2
      1
```

```
int n = 4;
for (int i=1; i<=n; i++){
    for (int j=1; j<=n-i; j++){
        Sop("-");
    }
    for (int j=1; j<=i; j++){
        Sop(j+" ");
    }
    System.out.println();
for (int i=n-1; i>=1; i--){
    for (int j=1; j<=n-i; j++){
        Sop("-");
    }
    for (int j=1; j<=i; j++){
        Sop(j+" ");
    }
    System.out.println();
}
```

Space = n - i
1 triangle = j<=i
2 triangle = j<=i
Outer 1 loop
i = 1 to n
Second outer loop:
i = n-1 to 1

Space = n - i
n - i + i = n
n - 1 × n = n × n n^2
$n^2 + n^2 = 2n^2$ Avoid Constants.
n^2

TC = $O(n^2)$
SC = $O(1)$

6)

```
1               1
1 2           2 1
1 2 3       3 2 1
1 2 3 4   3 2 1
1 2 3 4 3 2 1
1 2 3     3 2 1
1 2         2 1
1             1
```

space = 2*n-i
n = 4
triangle = j<=i

```
for (int i=1; j<=n; i++) {
    ~~for (int j=1; j<=2*n-i; j++){~~
        ~~Sop (" ");~~
    }
    for (int j=1; j<=i; j++){
        Sop (j);
    }
    for (int j=1; j<=2*(n-i); j++){
        Sop (" ");
    }
    for (int j=i; j>=1; j--){
        Sop (j);
    }
    System.out.println();
}
for (int i=n; i>=1; i--){
    for (int j=1; j<=i; j++){
        Sop (j);
    }
    for (int j=1; j<=2*(n-i); j++){
        Sop (" ");
    }
}
```

```
for (int j=n; j>=1; j--){
        Sop(j);
}
System.out.println();
```

TC :-

left- i = 1 Space = 6 right = 1 = 8
i = 2 Space = 4 right = 2 = 8
i = 3 2 3 = 8

for each row = n = 4 = 2n = 8
Same for lower part : = 2n

$2n \times 2n = 4n^2$

$O(4n^2)$ = Time Complexity
$O(1)$ = Space Complexity